A THEORETICAL EXPLORATION OF RSA CRYPTOGRAPHY

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Introduction: A Brief History of Cryptography

Cryptography is, put simply, the art of encoding and decoding information so that only specific, intended individuals can have access to it. Many different forms of cryptography have been developed and used by different cultures throughout the past several thousand years[1]. The simplest and earliest form of cryptography is the one time pad cipher, also known as the Vernam-cipher. This is, surprisingly, the only known method of cryptography that is mathematically unbreakable. Creating a Vernam-cipher is a simple matter of taking the words, letter, or symbols being used in a given message and replacing them with different, unrelated words, letter, or symbols[2]. The document that contains these symbol conversions is called the code pad, hence the name pad cipher.

Unfortunately, the one time pad cipher is only unbreakable for one use. Who could've guessed? Repeated use of the cipher could allow someone to break the code by cross referencing the contents of the message with real life events. To solve this re-usability issue, the Caesar cipher was developed. The Caesar cipher was a system commonly implemented by the Roman administration of Julius Caesar, though not developed by him, to securely communicate about matters of state and warfare during the first century B.C.[3]. It is implemented by shifting the alphabet, or set of symbols being used, by a specific number, and then using this as a guide for encoding and decoding a message[4]. A cipher pad for this system would look something like the following:

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A	B	C

This sort of cipher can be mathematically represented. If we let f be the encoding function and f^{-1} be the decoding function, where x is the numerical representation of the the letters of the alphabet(a=0, b=1,...) and n is the shift, then we have,

$$f(x) = x + n(\mod 26)$$

$$f^{-1}(x) = x - n(\mod 26).$$

The Caesar cipher is a marked improvement with regards to re-usability over the pad cipher; However, in the age of computers, it is trivial to break such a cipher. One need only decode the message with every possible shift. There are as many shifts as there are letters in the alphabet being used. The shift that makes the message make sense is probably the one used to encode it.

To solve the cryptographic issues of re-usability, decipherability, and efficiency, the RSA system, named after its three developers Ronald L. Rivest, Adi Shamir and Leonard M. Adleman, was introduced in the September 1977 edition of Scientific American[3]. The RSA system is a public key, asymmetric cryptography system designed to be implemented by computers that is easy to implement, extremely difficult to break, extremely reusable, and, given sufficient computing power and good implementation algorithms, very efficient[1]. The goal of this paper is to lay out a theoretical basis for the RSA system, explain its implementation, and outline its advantages.

Major Theorems and Results

The following are the theorems necessary for constructing a theoretical basis for the RSA system. The first four theorems will be offered without proof as they are either very well known results or fundamental theorems in the field of number theory. Theorems number 5 and 6 will be offered with proof as they are integral in the proof of Theorem 7. Theorem 7 will also be offered with proof as it is the crucial theorem upon which the RSA system hinges. The main source for the theorems and proofs presented in this section is [5].

Theorem 1. Let p be an integer such that $p \neq 1, \pm 1$, then p is prime iff $p|bc \implies p|b \vee p|c$, $\forall b, c \in \mathbb{Z}$.

Theorem 2. $a \equiv c(\mod n)$ iff $[a] = [c]$.

Theorem 3. If $p > 1$ is an integer, then the following are equivalent:

1.) p is prime.

2.) $\forall a \neq 0 \in \mathbb{Z}_p$, the equation $ax = 1$ has a solution in $\mathbb{Z}_p$.

3.) $bc = 0$ in $\mathbb{Z}_p \implies b = 0 \vee c = 0$.

Theorem 4 (Fermat's Little Theorem). If $a, p \in \mathbb{Z}$ with p prime, and $p \not| a$, then $a^{p-1} \equiv 1 (\mod p)$.

Theorem 5. Let a, n be integers where $n > 1$, then the equation $[a]x = [1]$ has a solution in $\mathbb{Z}_n$ iff $(a, n) = 1$ in $\mathbb{Z}$.

Proof of Theorem 5. Let a, n be integers where $n > 1$. Assume that $[w]$ is a solution to $[a]x = [1]$ in $\mathbb{Z}_n$. Then we have that,

$$[a][w] = [1]$$

$$\implies [aw] = [1]$$

$$\implies aw \equiv 1 (\mod n) \text{ in } \mathbb{Z}, \text{ by Theorem 2}$$

$$\implies aw - 1 = kn, \exists k \in \mathbb{Z}$$

$$\implies aw + n(-k) = 1.$$

Let d be an integer such that $(a, n) = d$. Thus, we know that $\exists r, s \in \mathbb{Z}$ such that $dr = a \wedge ds = n$. So, we have,

$$aw + n(-k) = 1$$

$$\implies drw + ds(-k) = 1$$

$$\implies d(rw - sk) = 1.$$

Thus we can say that $d|1$, and since d is positive by definition, we can say that $d = 1$. Thus, $(a, n) = 1$.

Now assume that $(a, n) = 1$ in $\mathbb{Z}$. Then, $\exists r, s \in \mathbb{Z}$ such that $ar + ns = 1$. So, we have,

$$ar + ns = 1$$

$$\implies ar - 1 = n(-s)$$

$$\implies ar \equiv 1 (\mod n)$$

$$\implies [ar] = 1 \text{ in } \mathbb{Z}_n, \text{ by Theorem 2,}$$

$$\implies [a][r] = [1].$$

Thus, $[r]$ is a solution to $[a]x = [1]$ in $\mathbb{Z}_n$.

Therefore, $[a]x = 1$ has a solution in $\mathbb{Z}_n$ iff $(a, n) = 1$ in $\mathbb{Z}$.

□

Theorem 6. If p and q are distinct prime numbers such that $p|c \wedge q|c, \exists c \in \mathbb{Z}$, then $pq|c$.

Proof of Theorem 6. Assume $p|c \wedge q|c$. Then,

$$c = pk, \exists k \in \mathbb{Z}$$

$$\implies q|pk$$

$$\implies q|p \vee q|k, \text{ by Theorem 1,}$$

$$\implies q|k$$

$$\implies k = ql, \exists l \in \mathbb{Z}$$

$$\implies c = pql$$

$$\implies pq|c.$$

Therefore, $p|c \wedge q|c \implies pq|c$.

□

Theorem 7. Let $p, q, n, k, e, d \in \mathbb{Z}$ such that:

1.)p and q are distinct positive primes.

2.) $n = pq$.

3.) $k = (p-1)(q-1)$.

4.) d is chosen so that $(d, k) = 1$.

5.) e is a solution to $dx \equiv 1(\mod k)$. We know this exists by Theorem 5.

Then $b^{ed} \equiv b(\mod n), \forall b \in \mathbb{Z}$.

Proof of Theorem 7. Since $de \equiv 1(\mod k)$ by definition, we can say that $de - 1 = kt, \exists t \in \mathbb{Z}$. Thus, $ed = kt + 1$. So, we have,

$$\begin{aligned} b^{ed} &= b^{kt+l} \\ &= b^{kt}b \\ &= b^{(p-1)(q-1)t}b \\ &= (b^{p-1})^{(q-1)t}b. \end{aligned}$$

Now we will proceed by cases.

Case 1: Assume $p \not| b$, then, by Fermat's Little Theorem:

$$b^{ed} = (b^{p-1})^{(q-1)t}b \equiv 1^{(q-1)t}b \equiv b(\mod p).$$

Case 2: Assume $p|b$. Then $b \equiv 0(\mod p) \implies b^{ed} \equiv 0(\mod p)$. Thus, $b^{ed} \equiv b(\mod p)$.

Thus, $b^{ed} \equiv b(\mod p)$. By a similar argument we can say that $b^{ed} \equiv b(\mod q)$. So, by definition,

$$p|b^{ed} - b \wedge q|b^{ed} - b.$$

So, by Theorem 6 we can say that, $pq|b^{ed}-b$. This is the same as $n|b^{ed}-b$. Therefore, $b^{ed} \equiv b(\mod n)$.

$\square$

Application

The RSA system is applied most commonly in encrypted computer networks such as secure wifi or bluetooth systems. Information pertaining to the implementation of RSA can be found in [6]. The implementation of the RSA system can be looked at as a 5 step process. The 5 steps are as follows:

1. Convert your Message:

The sender of a message first converts their message from whatever given language into a numerical representation. This is done by using what is essentially an extremely basic pad cipher. Here is an example:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

With this system of converting letters to numbers, we have 100 available characters to encode. The standard QWERTY keyboard has 94 possible characters on it, so we can send any message that can be typed in to a computer with this method.

Here we can see that if we let our message be 'MESSAGE'. Then the converted version of this message is $M = 13051919010705$.

2. Encode Your Message:

Once our message has been converted to a numerical representation, we will let M be the integer representing our complete message. Now we let $p, q, n, k, d, e \in \mathbb{Z}$ be the same as they are in Theorem 7. We choose p and q so that $M < pq = n$. We call our encoded message C. C is found be calculating the least residue modulo n of M^e.

3. Send Your Message:

We simply take our encoded message, C, and send it to our intended recipient in whatever way it pertinent.

4. Decode Your Message:

Having received the encoded message, the recipient must compute the least residue modulo n of C.

Recall that, by definition, $C \equiv M^e(\mod n)$. From this fact we can say that:

$$\begin{aligned} C &\equiv M^e(\mod n) \\ \implies C^d &\equiv M^{ed}(\mod n) \\ \implies C^d &\equiv M(\mod n), \text{ by Theorem 7.} \end{aligned}$$

Furthermore, because we have chose p and q so that $M < n$, we know that M must be the least residue modulo n of C^d. In this way, the recipient can retrieve the numerical representation of our message.

5. Convert Your Message(back):

Once the numerical representation, M, of the message has been retrieved, the recipient uses the exact same pad cipher to convert the message back into its linguistic representation.

Now that we have fully explored the theoretical basis and broader application strategy for the RSA system, we can consider what this would look like in practice.

As stated previously, RSA is typically implemented on secure networks of some sort, such as wifi or Bluetooth. Each computer on such a network would choose two relatively large prime numbers. These would be the values of p and q from Theorem 7. From these values, n, k, d, and e would then be calculated. These values can be identified or calculated rather straightforwardly by implementing certain algorithms on the computers in the network, such as the Chinese Remainder Theorem for example.

Once these values are determined, the values for e and n for each computer are published to the entire network. These are the 'public keys' which make RSA a public key cryptography system. With these values, any computer can encode messages and send them to any other computer on the network. Only the computer with the correct, unpublished decoding values can decipher messages encoded with their corresponding encoding values.

The only remaining issue in the system is that anyone on or eavesdropping on the network can send encoded messages to any computer on the network. This issue is circumvented through the implementation of digital signatures. Each computer on the network has a public identification key. When sending a message from computer 1 to computer 2, computer 1 would use its private decoding algorithm on its signature. Then computer 1 uses computer 2's encoding algorithm on its message as well as its 'decoded' signature. Computer 2 receives both the message and the signature. Computer 2 applies its decoding algorithm to both the signature and the message. It then applies computer 1's public encoding algorithm to the signature. Because the encoding and decoding algorithms are inverses of each other, applying both in any order is the same as applying the identity map. Thus, computer 2 retrieves the signature of computer 1 in this manner; furthermore, computer 2 knows that only computer 1 could've used its private decoding algorithm on its signature, so its identity is verified.

The security of the RSA system is dependent upon the problem of prime factorization. Given any integer m, we know from the Fundamental Theorem of Arithmetic that m can be expressed as a product of prime numbers. So, it is theoretically possible to take the published value n and break it down into its prime factors, p and q. Doing this would allow anyone to discover the secret decoding algorithms being used on a network, which would entirely invalidate the security of the network. However, this is essentially impossible in practice. Though it is possible to factor any integer into primes, there are no known algorithms efficient enough to do this in any practical setting[7]. If sufficiently large prime numbers are chosen for p and q, it could potentially take thousands of years to discover p and q from n[5].

Because of its relatively simple implementation, and its ability to reliably produce secure networks, the RSA encryption system is still widely used in cybersecurity and communications technology environments.

Bibliography

[1] Kotas, William August, "A Brief History of Cryptography" (2000). Chancellor' s Honors Program Projects.

[2]Priyam, Anuja. I. Extended Vigenère Using Double Transposition Cipher with One Time Pad Cipher — Semantic Scholar, www.semanticscholar.org/paper/I.-Extended-Vigen. Accessed 13 May 2024.

[3]CME's cryptology timeline. (n.d.). https://pi.math.cornell.edu/ morris/135/timeline.html

[4]Modified Caesar Cipher for Better Security Enhancement, www.researchgate.net/publication /269667378_Modified_Caesar_Cipher_for_Better_Security_Enhancement. Accessed 13 May 2024.

[5] Hungerford, Thomas W. Abstract Algebra: An Introduction. Cengage, 2020.

[6] Zwillinger, D. (2018). CRC standard mathematical tables and formulas. Boca Raton: CRC Press, Taylor & Francis Group.

[7] Wang, B., Hu, F., Yao, H., & Wang, C. (2020). Prime factorization algorithm based on parameter optimization of Ising model. Scientific Reports, 10(1). https://doi.org/10.1038/s41598-020-62802-5

On an entirely unrelated note, thank you for teaching me so much math. Even though your classes are extremely difficult, I can confidently say that I have gained more knowledge of mathematics from three of your classes than I have from every other math course I have ever taken. I appreciate the rigor, though I did complain about it frequently, and I appreciate you teaching me. -Sean Meehan